5. Audio coding standards

based in part from on Spanias, Painter & Atti, Chapter 10

Key question: how do standard codecs combine transforms, psychoacoustics, quantization and bitstream syntax?

Roadmap

From efficient waveform compression to adaptive, immersive audio

01 · Representations

MIDI · mono/stereo/surround · channel-based audio

02 · Perceptual coding

MPEG standards · MP3 · AAC · MPEG-4 Audio

03 · Modern coding

Opus · xHE-AAC · speech/music · low latency

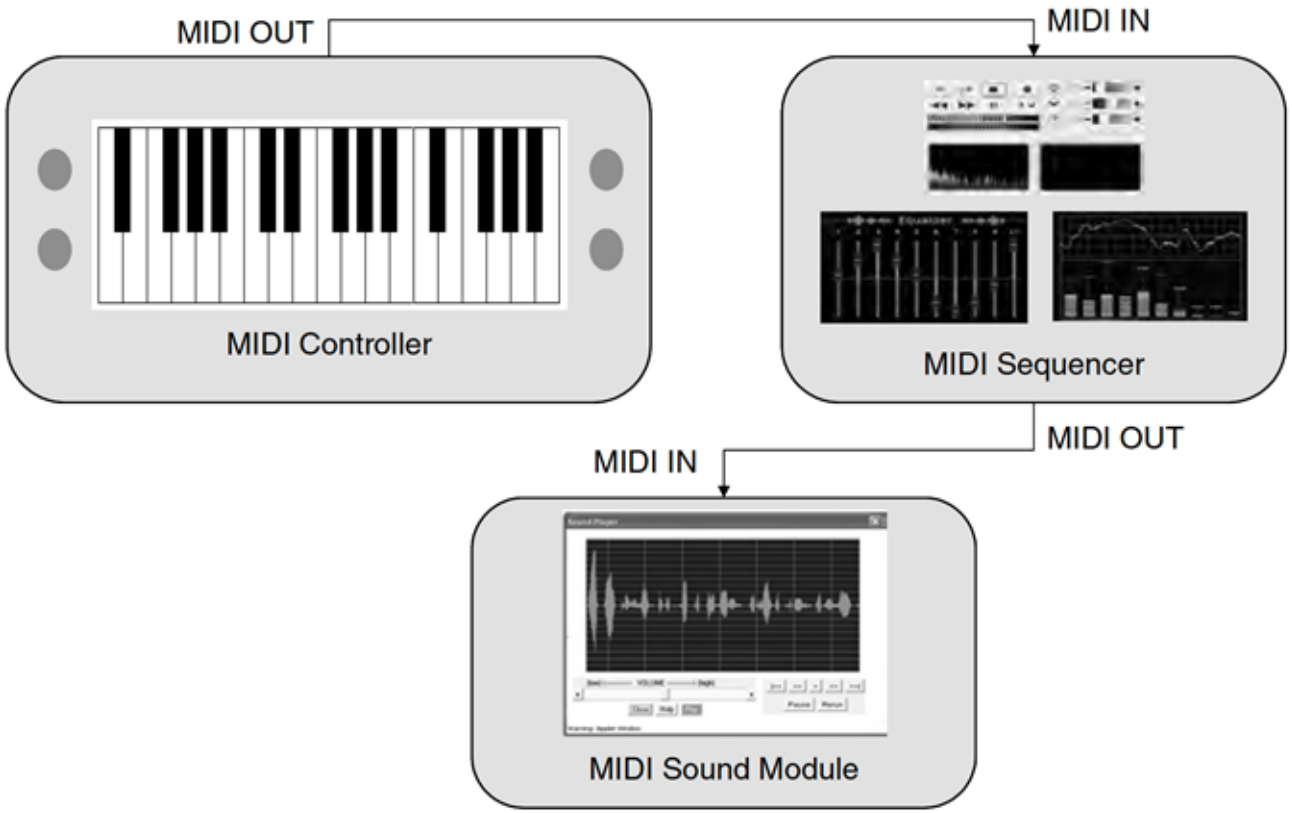
04 · Immersive + lossless

Dolby Atmos · JOC · MPEG-H · FLAC · FFmpeg

Key distinction Compressing waveforms ≠ representing and rendering an audio scene

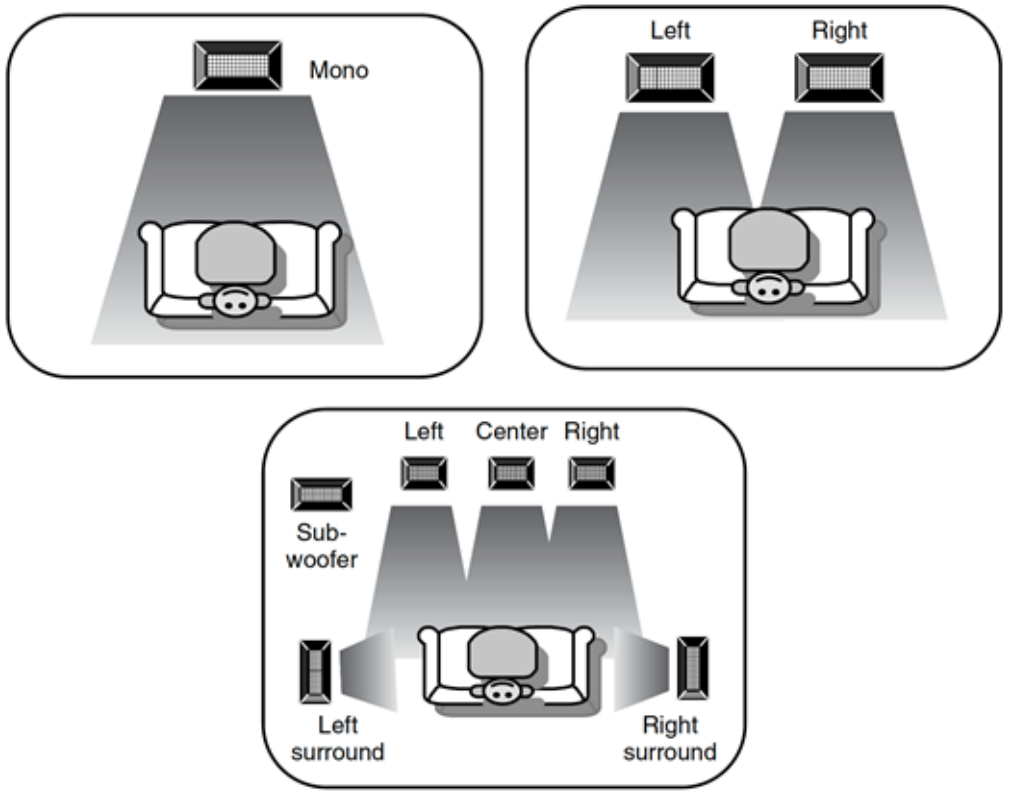
MIDI: symbolic control, not audio samples

MIDI transmits performance events and control messages between devices.



Mono, stereo and surround formats

Channel layout determines how many audio streams and spatial cues must be preserved.



MPEG audio standards: the first generations

From stereo perceptual coding to efficient multichannel audio.

Standard	Main innovation	Why it matters
MPEG-1 Audio 1992	Three layers of perceptual audio coding (I, II and III).	Layer III became MP3; mono and stereo audio.
MPEG-2 Audio 1994	Extends MPEG-1: lower sampling rates and backward-compatible multichannel audio.	Early surround sound and digital broadcasting.
MPEG-2 AAC 1997	A new, more efficient perceptual coder, designed for multichannel audio.	AAC; not backward-compatible with MPEG-1 decoders.

KEY POINT MPEG-1 and MPEG-2 established the core foundations of modern perceptual audio coding.

MPEG standards: audio, metadata and frameworks

These standards address different parts of the multimedia ecosystem.

Standard	Main purpose	Role
MPEG-4 Audio **1998 onward**	A toolbox for audio: AAC, speech, scalable, parametric and synthetic coding.	Codes and represents audio.
MPEG-7 **2001**	Standardizes metadata describing multimedia content.	Search and indexing; not an audio codec.
MPEG-21 **2002 onward**	A framework for identifying, managing and exchanging multimedia resources.	Interoperability and rights; not an audio codec.

KEY POINT MPEG-4 includes audio codecs; MPEG-7 and MPEG-21 do not compress audio.

MPEG-1 Audio

Lossy compression for mono and stereo signals.

- Sampling frequencies: 32 kHz, 44.1 kHz and 48 kHz.
- Precision: 16 bits.
- Three compression layers:
  - Layer 1: 32–448 kb/s, target 192 kb/s
  - Layer 2: 32–384 kb/s, target 128 kb/s
  - Layer 3: 32–320 kb/s, target 64 kb/s
- Layer 3 is the MP3 format, widely used for Internet applications.

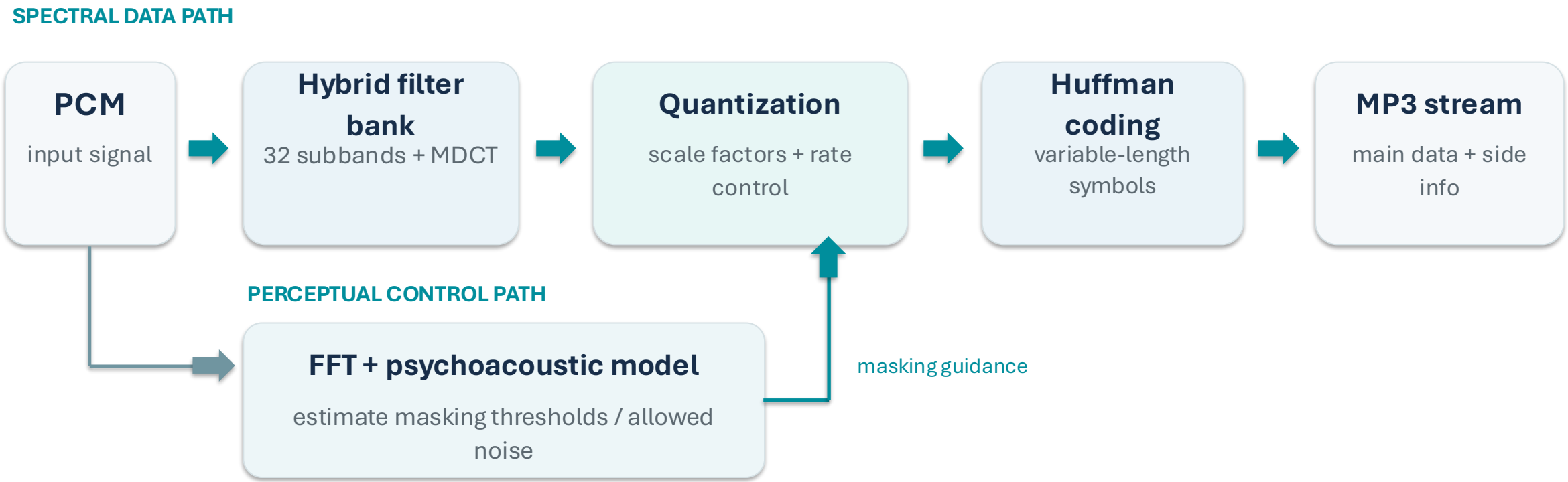
Layer 3 = MP3

Same family, increasing compression complexity from Layer 1 to Layer 3.

MP3 IN PRACTICE Built into all mobiles, computers, and used by all music apps.

MP3 encoder: two cooperating paths

Hybrid analysis creates coefficients; psychoacoustics tells the encoder where distortion is tolerable.



KEY IDEA Bit allocation balances audible quantization noise and the available bitrate.

MP3 encoder: perceptual versus rate control

Two nested decisions determine how accurately the spectral coefficients are coded.

Perceptual guidance FFT + critical bands → masking thresholds in each frequency region

OUTER LOOP

Control audible distortion

Compare quantization noise with masking thresholds; adjust scale factors by frequency region.

INNER LOOP

Meet the bit budget

Quantize and Huffman-code; adjust global gain until the coded data fits the available bits.

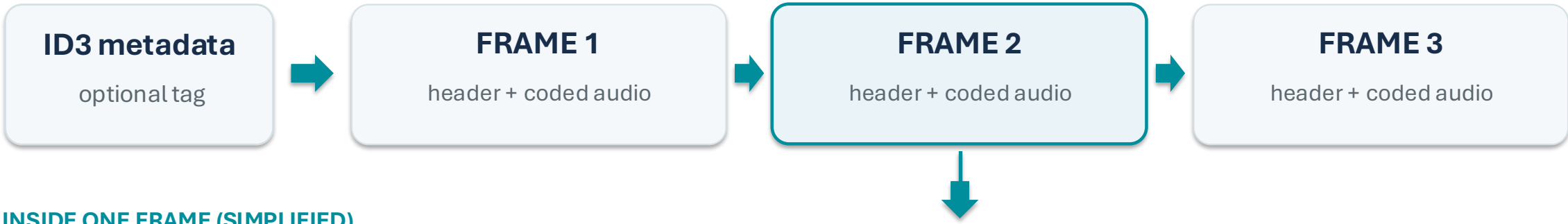
iterate until both conditions are acceptable

TAKEAWAY The MP3 bitstream is standardized; encoder decisions can differ.

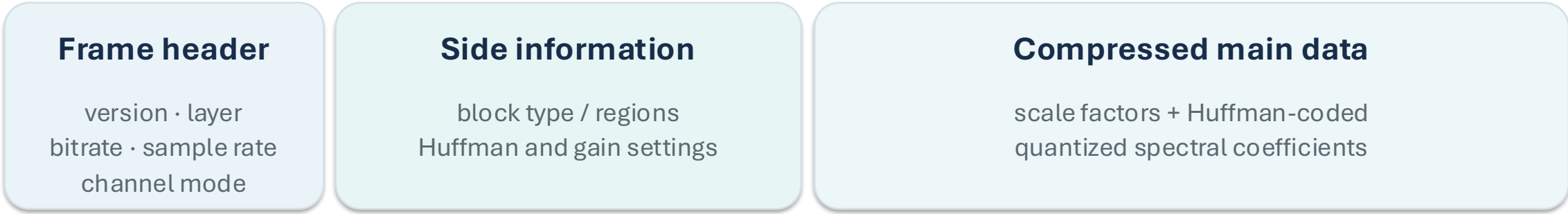
MP3 bitstream: frames and their contents

An MP3 file contains successive audio frames, optionally accompanied by file-level metadata.

FILE STRUCTURE



INSIDE ONE FRAME (SIMPLIFIED)



Simplified layout: optional CRC and ancillary data are not shown.

REMEMBER A bit reservoir can share coded-data capacity between neighboring frames.

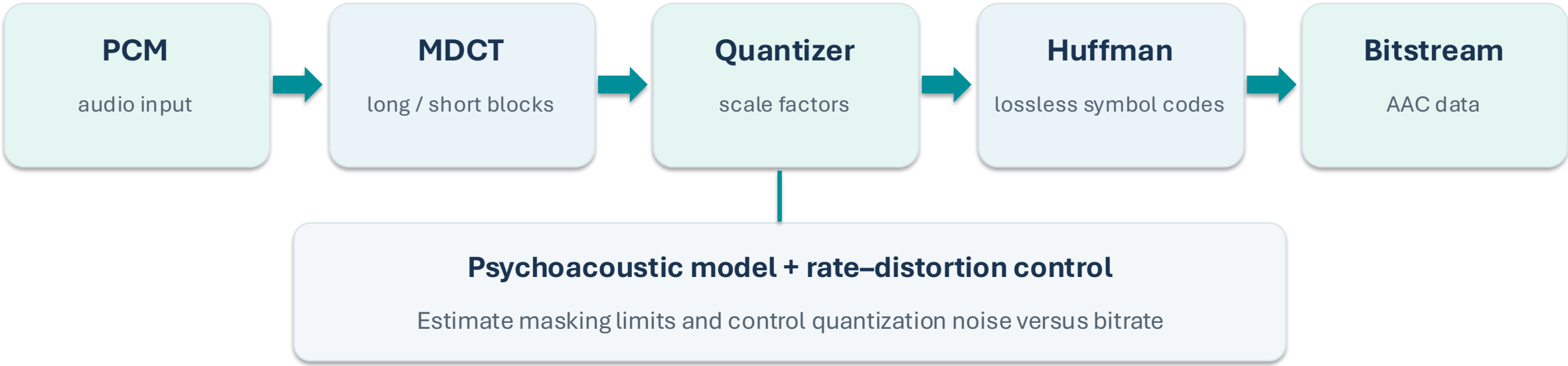
MPEG-2 Advanced Audio Coding (AAC)

AAC follows the Layer-3 paradigm but adds tools for improved low-bitrate quality.

- Supports mono/stereo and multichannel audio (up to 5.1 in MPEG-2 AAC).
- Transform-based perceptual coding: MDCT, non-uniform quantization, Huffman coding and rate–distortion control.
- Compared with MP3: higher frequency resolution (1024 lines), improved joint stereo and block switching.
- Additional tools include temporal noise shaping, prediction and channel coupling.

AAC-LC: the core perceptual coding chain

The widely deployed AAC-LC profile uses MDCT-based analysis rather than the MP3 hybrid filter bank.



AAC: transform coefficients + perceptual bit allocation

Compared with MP3: new filter-bank and coding tools

AAC-LC: tools that improve perceptual coding

Some tools are adapted to the signal; AAC-LC is a profile within the larger MPEG-4 Audio toolbox.

Block switching

Long MDCT windows resolve steady tones; short windows reduce temporal smearing around attacks.

Temporal Noise Shaping (TNS)

Controls how quantization noise is distributed in time, especially around transients.

Joint stereo: M/S and intensity

Shares information between left and right channels when independent coding is wasteful.

Quantization and entropy coding

Uses scale factors, rate–distortion decisions and Huffman coding of spectral values.

Important distinction: prediction is present in some AAC profiles, but not in the common AAC-LC profile.

AAC: developed in MPEG-2, adopted in MPEG-4

AAC has an MPEG-2 origin, but its most familiar implementations belong to MPEG-4 Audio.

MPEG-2 AAC

1997 · ISO/IEC 13818-7

Introduces the new AAC perceptual codec for mono, stereo and multichannel audio.

MPEG-4 AAC

ISO/IEC 14496-3

Carries AAC forward and defines audio object types and profiles, including AAC-LC.

AAC family

HE-AAC and xHE-AAC

Adds new delivery choices, from low-bitrate extensions to the USAC-based profile.

REMEMBER The standard in which AAC first appeared is not the only standard where AAC is defined and used.

Why most AAC is discussed as MPEG-4 Audio

MPEG-4 Audio organizes AAC into profiles; the MP4/M4A format is a container, not a codec.

STANDARD

MPEG-4 Audio

ISO/IEC 14496-3
Defines audio object types, profiles and a decoder bitstream syntax.

CODEC / PROFILE

AAC-LC, HE-AAC

Different members of the MPEG-4 AAC family serve different bitrate and quality needs.

FILE CONTAINER

.m4a / .mp4

An MP4-family file can carry AAC audio. The filename does not specify the AAC profile.

IN PRACTICE AAC music files (.m4a), AAC in .mp4 video, and AAC delivery in web streaming. Spotify, Youtube.

The AAC family: extensions versus a new coding core

HE-AAC extends AAC-LC; xHE-AAC introduces USAC while preserving AAC-family decoder support.

AAC-LC-BASED EXTENSIONS

AAC-LC

MDCT-based
core codec

→

HE-AAC v1

AAC-LC + SBR
(high frequencies)

→

HE-AAC v2

HE-AAC v1 + PS
(stereo cues)

USAC: A NEW CORE (MPEG-D)

USAC

Speech prediction + transform audio

→

xHE-AAC (Extended HE-AAC profile)

Uses USAC for new streams; decoder profile also covers older AAC tools

KEY IDEA xHE-AAC is not made simply by adding one more tool to an AAC-LC bitstream.

HE-AAC: efficient delivery at low bitrates

Two MPEG-4 AAC profiles extend AAC-LC with parametric reconstruction tools.

AAC-LC
core audio

→

HE-AAC v1
+ SBR

→

HE-AAC v2
+ Parametric Stereo

Spectral Band Replication (SBR)

Transmit a coded lower band and compact information for high-frequency reconstruction.

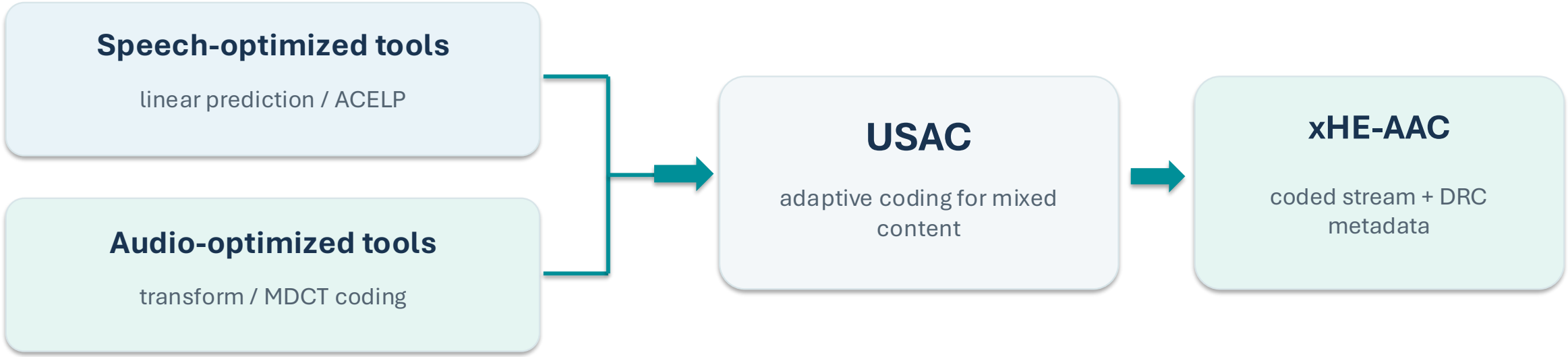
Parametric Stereo (PS)

At very low rates, encode shared audio plus compact cues for reconstructing stereo.

IN PRACTICE DAB+ (Digital Audio Broadcasting Plus), used in terrestrial radio networks

xHE-AAC: USAC for speech, music and streaming

New USAC coding architecture (MPEG-D, Part 3) + integrated loudness / dynamic-range control.



Wide operating bitrate range

Useful for adaptive delivery and speech/music mixes.

Compatibility: profile versus bitstream

Legacy AAC support does not make USAC an AAC-LC extension.

IN PRACTICE Netflix, Instagram, Facebook, Android, Apple, Xbox, DRM.

Opus: one codec for speech and music

IETF RFC 6716 · especially relevant to interactive communication and low latency.

SILK

Predictive / SILK

Linear-prediction-based coding, optimized for speech.

CELT

Transform / CELT

MDCT-based coding, suited to music and low delay.



Adaptive operating modes

- SILK-only
- CELT-only
- Hybrid SILK + CELT

Practical traits: variable bitrate · adaptable frame duration · packet-loss concealment · WebRTC

OPUS IN PRACTICE Discord voice chat · WebRTC calls in Chrome / Firefox / Safari

Audio codecs: different optimization goals

Compare coding principle, low-bitrate efficiency, latency and deployment—not one universal winner.

Codec	Core strategy	Strength	Example uses
MP3	hybrid transform	legacy compatibility	downloads / archives
AAC-LC	MDCT perceptual	general-purpose audio	video / streaming
HE-AAC v1/v2	AAC-LC + SBR / PS	efficient low-rate stereo	digital radio / streaming
xHE-AAC	USAC: LP + transform	very low to high rates	adaptive delivery
Opus	SILK / CELT / hybrid	low delay + versatility	calls / WebRTC

Dolby AC-3 (Dolby Digital)

A de facto standard for movie theaters and home theater systems.

- Sample rates: 32, 44.1, and 48 kHz
- Bit rates: 32–640 kb/s, variable
- High-quality output at 64 kb/s per channel
- Delay roughly 100 ms
- MDCT filter bank
- Multiple channels processed as an ensemble
- Robust decoder downmix functionality from 5.1 to fewer channels

AC-3 was designed around multichannel playback and robust consumer decoding.

AC-3 IN PRACTICE DVD-Video surround tracks · digital TV and home-theater receivers

Dolby Digital (AC-3): coding fixed channels

A conventional multichannel codec: the decoder reconstructs a 5.1 channel mix.

PCM 5.1
six fixed channels

→

MDCT analysis
window / transforms

→

Perceptual model
masking + bit allocation

→

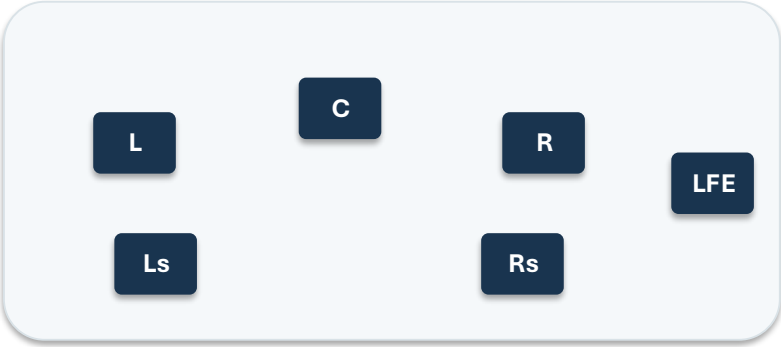
Quantization
coded spectral coefficients

→

AC-3 stream
frames for decoding

The result remains channel-based

After decoding, the six channel signals are routed or downmixed for the playback system. Object positions are not transmitted.



Fixed speaker channels

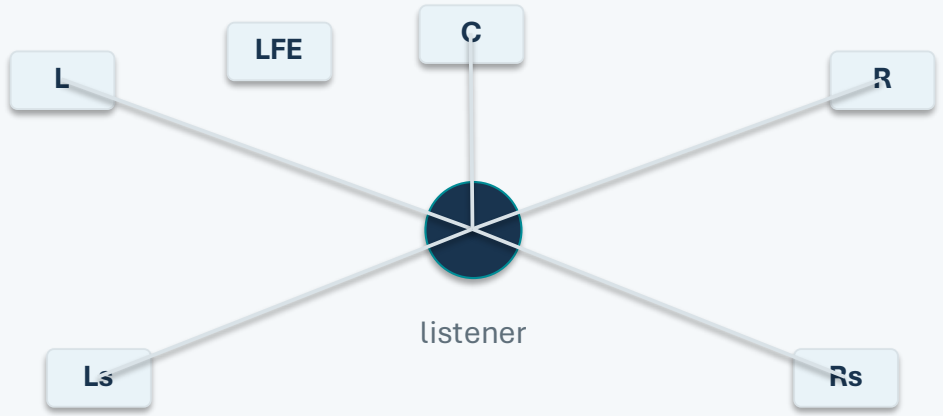
KEY IDEA AC-3 codes channel signals; immersive systems also represent a spatial sound scene.

From channel-based to object-based audio

The central change is what the mix describes—not just how many channels it contains.

CHANNEL-BASED · 5.1

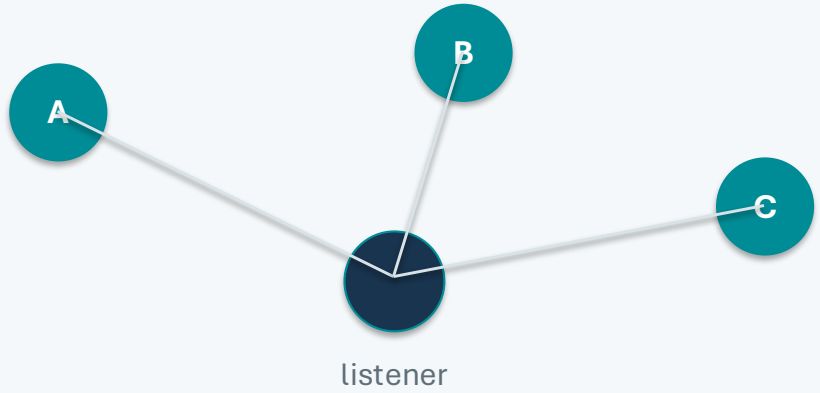
Sound is assigned to fixed channels.



Same set of output channels in the encoded mix

OBJECT-BASED · 3-D SCENE

Sound sources carry spatial metadata.



Position (x, y, z) can vary with time

KEY IDEA Objects are placed by a renderer; fixed-channel beds can still coexist with them.

Dolby Atmos: beds, objects and metadata

An Atmos production can combine fixed-channel material with independent spatial sound sources.

Channel bed

Conventional audio channels (up to 7.1.2 in production).

Audio objects

Independent signals for selected sounds.

Spatial metadata

Object positions and movement over time.



Atmos scene / spatial renderer

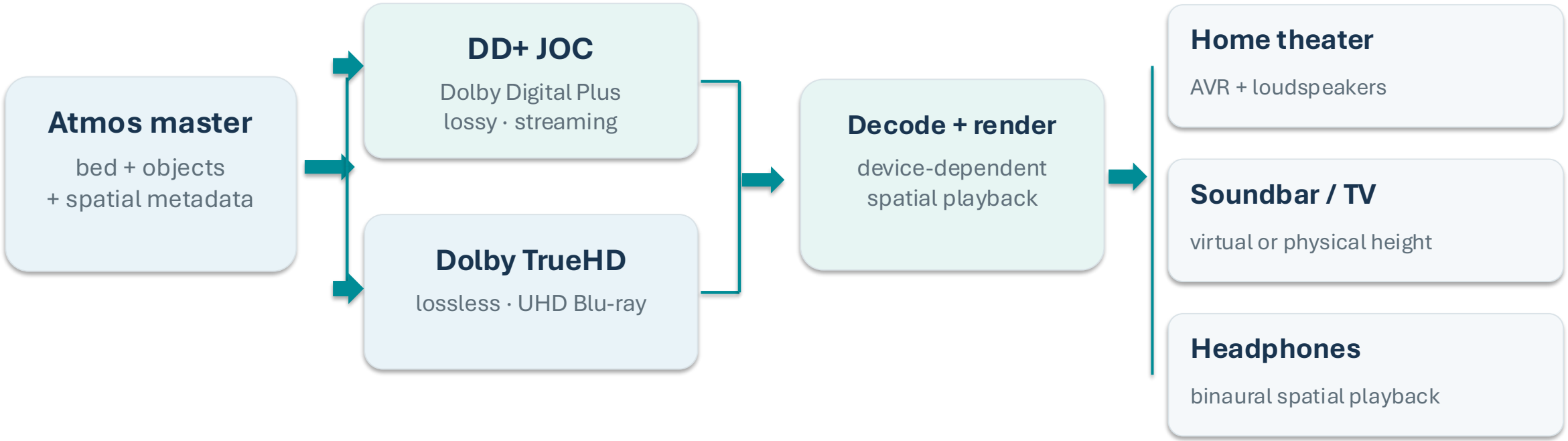
Maps the mix to an actual loudspeaker or headphone layout

Production may contain up to 128 audio inputs; the number delivered depends on the encoding format.

KEY IDEA Height channels and movable objects are different concepts.

Dolby Atmos: scene, delivery and reproduction

The immersive mix, distribution codec and final playback rendering are separate stages.



Other immersive delivery technologies include Dolby AC-4; cinema has a separate Atmos distribution workflow.

KEY IDEA Dolby Atmos is not synonymous with a single compression codec.

Atmos: from production to playback

Rendering happens according to the available speakers or headphone configuration.

Mix / master

beds + objects
+ metadata

Distribution examples

- DD+ JOC: streaming (lossy)
- TrueHD: Blu-ray (lossless)
- AC-4: selected delivery systems

Decoder + renderer

spatial mapping depends on the device

Atmos describes the scene; transport codecs determine how efficiently it is delivered.

Cinema large speaker arrays

Home soundbar / AVR

Headphones binaural rendering

Dolby Digital Plus JOC: efficient Atmos delivery

Joint Object Coding provides backward-compatible object-based playback.

Atmos production mix

beds, objects,
spatial metadata

DD+ JOC bitstream

multichannel render
+ JOC side information

Regular DD+ decoder

outputs channel mix

Atmos-aware decoder

JOC reconstruction +
metadata for renderer

Important distinction

JOC delivers up to 15 full-range channels/objects plus LFE, rather than separately coding every production object.

MPEG-H 3D Audio: immersive and interactive

ISO/IEC 23008-3 · standardized audio coding, scene description and reproduction.

Channels

conventional beds

Objects

signals + metadata

HOA

sound-field representation

Personalization under producer control

Choose a language or commentator · boost dialogue · render for speakers, soundbars or headphones

MPEG-H IN PRACTICE South Korean ATSC 3.0 television · Sony 360 Reality Audio format

Immersive audio: compare four distinct layers

Do not confuse how audio is represented, encoded, decoded and rendered.

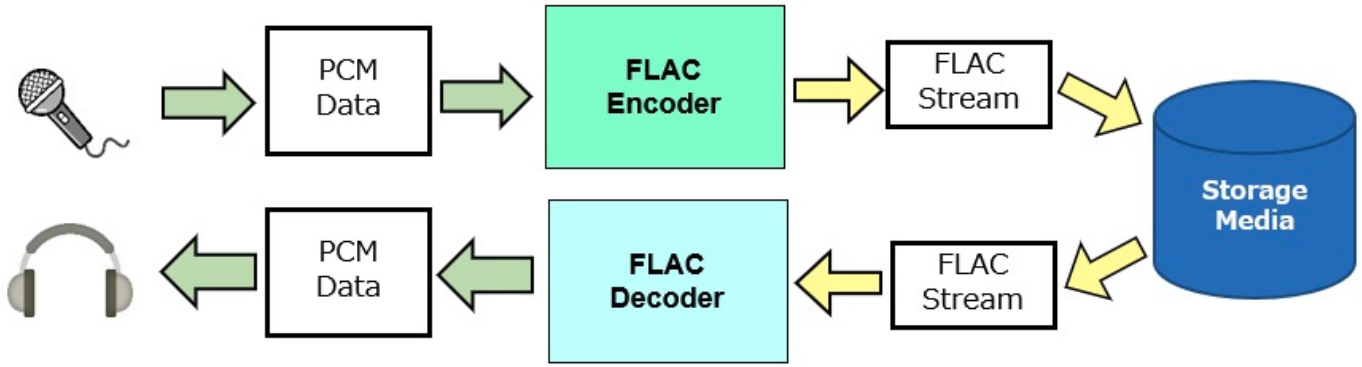
	Conventional 5.1	Dolby Atmos	MPEG-H 3D Audio
Representation	Fixed output channels	Channel beds + positioned objects	Channels + objects + HOA
Coding / delivery	AC-3, DD+ or other channel codecs	e.g. DD+ JOC (lossy) TrueHD (lossless)	MPEG-H 3D Audio bitstream
Playback	Channel mapping + downmix	Renderer adapts to speaker configuration	Renderer adapts to reproduction setup
Personalization	Limited by mix and application	Application / format dependent	Language and dialogue choices supported

TO REMEMBER Representation, codec and renderer solve different problems.

Lossless audio coding: FLAC

FLAC preserves the original PCM audio exactly while reducing file size.

- Lossless compression: decoded PCM is bit-for-bit identical to the original.
- Typically compresses music substantially, depending on the signal.
- Open, non-proprietary format with fast decoding and rich metadata.
- Real products: TIDAL High/Max FLAC streaming and Bandcamp FLAC downloads.



FLAC properties

- **Lossless:** The encoding of audio (PCM) data incurs no loss of information, and the decoded audio is bit-for-bit identical to what went into the encoder.
- **Fast:** FLAC is asymmetric in favor of decode speed. Decoding requires only integer arithmetic, and is much less compute-intensive than for most perceptual codecs.
- **Hardware support:** FLAC is supported by dozens of consumer electronic devices, from portable players, to home stereo equipment, to car stereo.
- **Flexible metadata:** FLAC's metadata system supports tags, cover art, seek tables, and cue sheets. Applications can write their own APPLICATION metadata once they register an ID.
- **Seekable:** FLAC supports fast sample-accurate seeking. Not only is this useful for playback, it makes FLAC files suitable for use in editing applications.
- **Streamable:** Each FLAC frame contains enough data to decode that frame.

FFmpeg: experiment with codec choices

An open-source toolkit for transcoding, probing and evaluating encoded audio.

Opus / interactive audio

```
ffmpeg -i input.wav -c:a libopus -b:a 96k output.opus
```

AAC / general-purpose audio

```
ffmpeg -i input.wav -c:a aac -b:a 192k output.m4a
```

FLAC / lossless

```
ffmpeg -i input.wav -c:a flac output.flac
```

What to remember

Audio coding now optimizes more than file size.

01 · Signal coding

MP3 and AAC compress PCM using perceptual models, transforms, quantization and entropy coding.

02 · Application needs

Opus targets low delay and robustness; xHE-AAC spans speech and music at diverse bitrates.

03 · Sound scenes

Atmos and MPEG-H describe spatial information; renderers adapt playback to the listener.

Main distinction: codec • spatial representation • rendering • delivery format

References and further reading

Spanias, Painter & Atti (2007), Audio Signal Processing and Coding

Dolby — Beds and Objects in Dolby Atmos

IETF RFC 6716 — Opus Audio Codec

Dolby — Dolby Digital Plus JOC

MPEG-D USAC (xHE-AAC) — ISO/IEC 23003-3

Fraunhofer IIS — MPEG-H Audio

FFmpeg documentation

Free Lossless Audio Codec (FLAC)